Project - Bond Analytics Dashboard

Overview:

This project involves creating a visualiser screen to explore how a bond's price is determined by its cashflows and, in turn, how that determines its duration(s) and DV01 given an interest rate (curve). Aiming for an audience of finance students with a good grounding in maths, this visualiser should give a good intuition of how a given bond's price is determined and should let you explore how its yield to maturity and price change as interest rates change.

Alongside this, a brief presentation (up to 4 slides) should be created to explain what the screen does and add any extra details about the figures and plots it displays.

Key Outcomes:

- Visual & interactive demonstration of relationship between YTM and price, and how these are affected by interest rates.
- Calculation of Macaulay duration, modified duration/volatility, DV01, again with plots or other interactive elements to help build understanding.
- Explore loading bond data from a simple API source such as yfinance
- A brief presentation with annotations/a voice explainer on what you've created and how, with some background on both the finance theory, maths and programming decisions.
    - On the presentation, explain how using (i.e. buying/selling) bonds of a certain duration can hedge interest rate risks of bonds of other durations, using visualisations from your app.

Technical Pointers

The dashboard should be a simple, single screen webapp, running locally, written entirely in Python, using a library such as *Streamlit* or *Plotly Dash*. For the 'backend', *Polars* is the recommended approach. Share your code via GitHub. Focus on applying good design principles to make an intuitive UI. AI use isn't banned but be ready to explain any line/function in your code and any design decisions you've made.

Resources

This lecture series (up to the end of *Fixed Income Securities*) is great for building intuition. Also try Brealey, Myers and Allen's *Principles of Corporate Finance*.